

DHRUV POONIA

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PROFESSIONAL SUMMARY

Computer Science undergraduate (graduating May 2027) with hands-on experience in full-stack and Android development, AI/ML, and computer vision. Proven ability to build end-to-end solutions — from a production-deployed Android app at a PSU internship to deep learning pipelines benchmarking multiple CNN architectures. AWS Certified Machine Learning Engineer with skills spanning Python, Java, React.js, OpenCV, and YOLO. Seeking software engineering or ML engineering roles where I can contribute to impactful, real-world systems.

EDUCATION

SRM Institute of Science and Technology — Kattankulathur, Tamil Nadu
B.Tech, Computer Science and Engineering

May 2023 – May 2027

EXPERIENCE

Rajasthan Electronics & Instruments Limited (REIL) — Jaipur, Rajasthan
Software Developer Intern

Dec 2025 – Feb 2026

- Engineered an Android-based Electronic Milk Analyzer (EMA) management application to **digitize device tracking and calibration workflows**, replacing paper-based logs and reducing manual record-keeping across multiple regional service locations.
- Implemented QR code-based device identification, calibration report generation, and maintenance history tracking, **consolidating three previously fragmented manual workflows** into a single mobile interface and improving service traceability for field engineers.
- Built the application end-to-end using Java, XML, and SQLite with an **offline-first architecture**, designing a local data model and form-based capture UI that ensured reliable data logging in low-connectivity, on-site field environments.

PROJECTS

Smart Helmet Detection System

2025

Python, OpenCV, YOLO · Computer Vision

- Built a **real-time helmet detection system** using YOLO and OpenCV that processes live video feeds frame-by-frame, automatically flagging riders without protective helmets to support traffic-safety enforcement and large-scale surveillance applications.
- Tuned the YOLO inference pipeline for **low-latency, frame-by-frame video processing**, optimising model performance to support deployment-ready monitoring under realistic traffic and surveillance conditions.

Driver Monitoring System

2025

Python, OpenCV, Web Stack, Database · Full-Stack + AI

- Developed a **full-stack driver drowsiness detection system** combining computer-vision-based eye and facial-state tracking with a web frontend, backend service, and database for continuous, real-time monitoring of driver alertness during active operation.
- Integrated a **Telegram bot** to deliver **instant safety notifications** upon detection of drowsy or inattentive driving, enabling supervisors or family contacts to intervene within seconds of a flagged drowsiness event.

Cotton Disease Detection

2025

Python, CNN, VGG16, DenseNet, EfficientNet · Deep Learning

- Built a **multi-model deep-learning pipeline** combining cotton-leaf imagery with environmental data to classify disease categories and surface actionable crop-health insights aimed at early intervention and yield protection for farmers.
- Benchmarked four CNN architectures** (custom CNN, VGG16, DenseNet, EfficientNet) on the same dataset to compare classification accuracy, model footprint, and inference cost, identifying the most viable trade-off for low-resource agricultural deployment.

Travel Sphere

Aug 2024 – Oct 2024

Java, Python, HTML, CSS · Application Development

- Designed and developed a multi-feature travel application supporting hotel booking, itinerary planning, map integration, and local-attraction discovery, unifying four interconnected modules into a single **end-to-end trip-planning interface**.
- Built the application using a **Java + Python backend** paired with HTML/CSS frontend screens, modelling user journeys from initial destination search through booking confirmation and on-trip navigation.

LEADERSHIP & VOLUNTEERING

Committee Head — AARUUSH, SRM University

Jun 2025 – Nov 2025

- Led **end-to-end planning and on-ground execution** of multiple university-wide events under AARUUSH, coordinating logistics, scheduling, and resource allocation across cross-functional student teams over a **six-month build-up cycle**.
- Directed **creative design and promotional initiatives** end-to-end — from concept and visual collateral through campaign rollout — driving event participation and strengthening AARUUSH's on-campus and digital visibility.

Volunteer — RIHR NGO

Jun 2025 – Jul 2025

- Analysed and compiled **multiple raw datasets** on juvenile crime and child welfare, cleaning inconsistent records and restructuring fragmented data into a clean, **analysis-ready format** suitable for downstream reporting and stakeholder review.
- Authored a **consolidated analytical report** — subsequently **published on the NGO's website** — that translated dataset findings into accessible insights to strengthen the organisation's awareness, outreach, and community advocacy efforts.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, SQL, JavaScript

Web Development: HTML, CSS, React.js, Next.js, Node.js, REST APIs

App Development: Android Studio (Java / Kotlin), XML, Flutter

AI / ML: Machine Learning, Deep Learning, Computer Vision, OpenCV, YOLO, TensorFlow, PyTorch, scikit-learn, Pandas, NumPy

Tools & Platforms: Git & GitHub, VS Code, Android Studio, MySQL, Firebase, MATLAB, AWS

Spoken Languages: English (Full Professional) · Hindi (Native) · French (Beginner)

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Machine Learning Engineer – Associate · Amazon Web Services, 2026
- NPTEL — Supervised Machine Learning · Introduction to Machine Learning · Java Programming (2024 – 2025)
- MATLAB Onramp & Image Processing
- Udemy – C++ course
- Workshops & Events —IBR, Aaruush · SRM Hackathon 8.0 · UI/UX, Cherry+ (2024)